MINING DEALER BEST PRACTICE

3500 Oil Pump Lifting Tool

Maintenance and Repair | Application | Component Life Management | **Component Renewal (CRC)** | MARC Management

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1.0 Introduction

During the assembly process of a 3500 engine rebuild, proper installation of major components is critical. It is essential to take care of the components such as the oil pump to ensure optimal system operation.

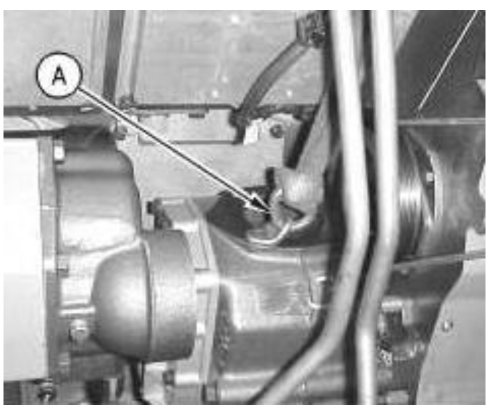
Currently, there is no assigned tool to lift the oil pump during installation. This introduces various risk factors and makes the process dependent on each specific technician and their experience level. Therefore, a lifting tool was designed to facilitate this process, as well as help with the alignment of the pump with the drive in the front housing of the engine.

This best practice aims to improve the manipulation of oil pumps during their installation, while decreasing the probability of safety injuries such as blows and scratches to technicians' hands.

2.0 Best Practice Description

Information on the oil pump lifting tool designed is as follows:

- Geometry: Diagonally bent plate
- Appearance: Similar to an engine lifting eyebolt
- Dimensions: 120mm depth, 120mm width, 245mm height
- Weight: 1.67kgs
- Materials: ASTM A36 ¼"steel sheet
- Structure: A lifting eyebolt plate welded to a similar plate of equal thickness with a 3/8" clearance hole
- Personnel Involved (Roles): One welder for the entire manufacturing process



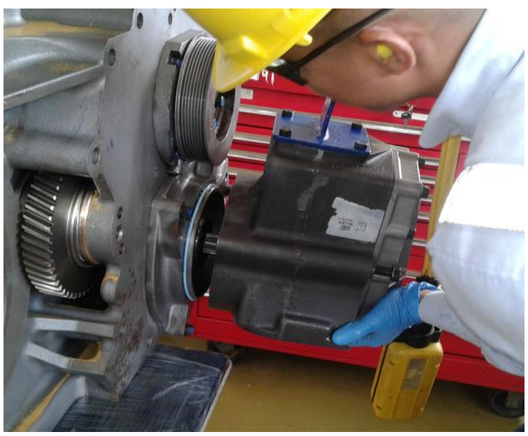
Before

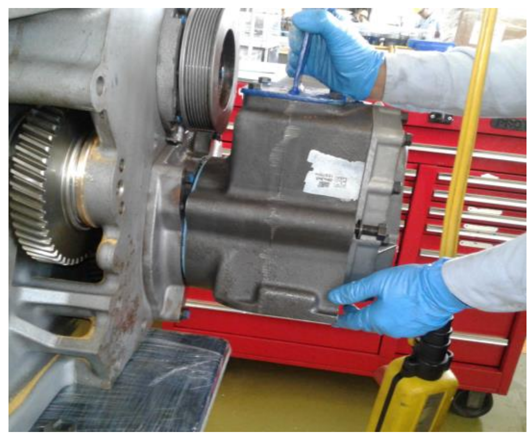
After

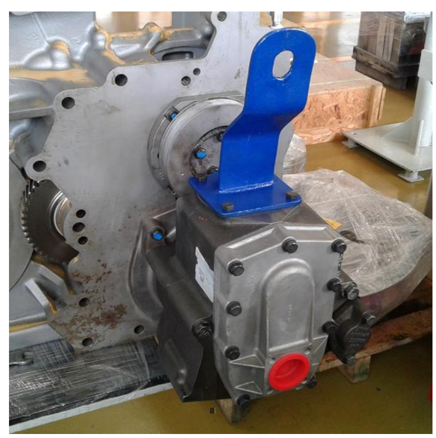
3.0 Implementation Steps

Step 1: Safe transportation of the oil pump to the engine has to be taken into account

Step 2: Design on lifting tool has special angles to keep the center of gravity leveled toward the pinion-drive teeth







4.0 Benefits

The benefits offered by this improvement are as follows:

- Improved grip of the component during transportation; thus, reducing safety risks for the technician carrying the 45-kg oil pump
- Improved pump maneuverability during installation when having to align with the drive in the front of the engine housing

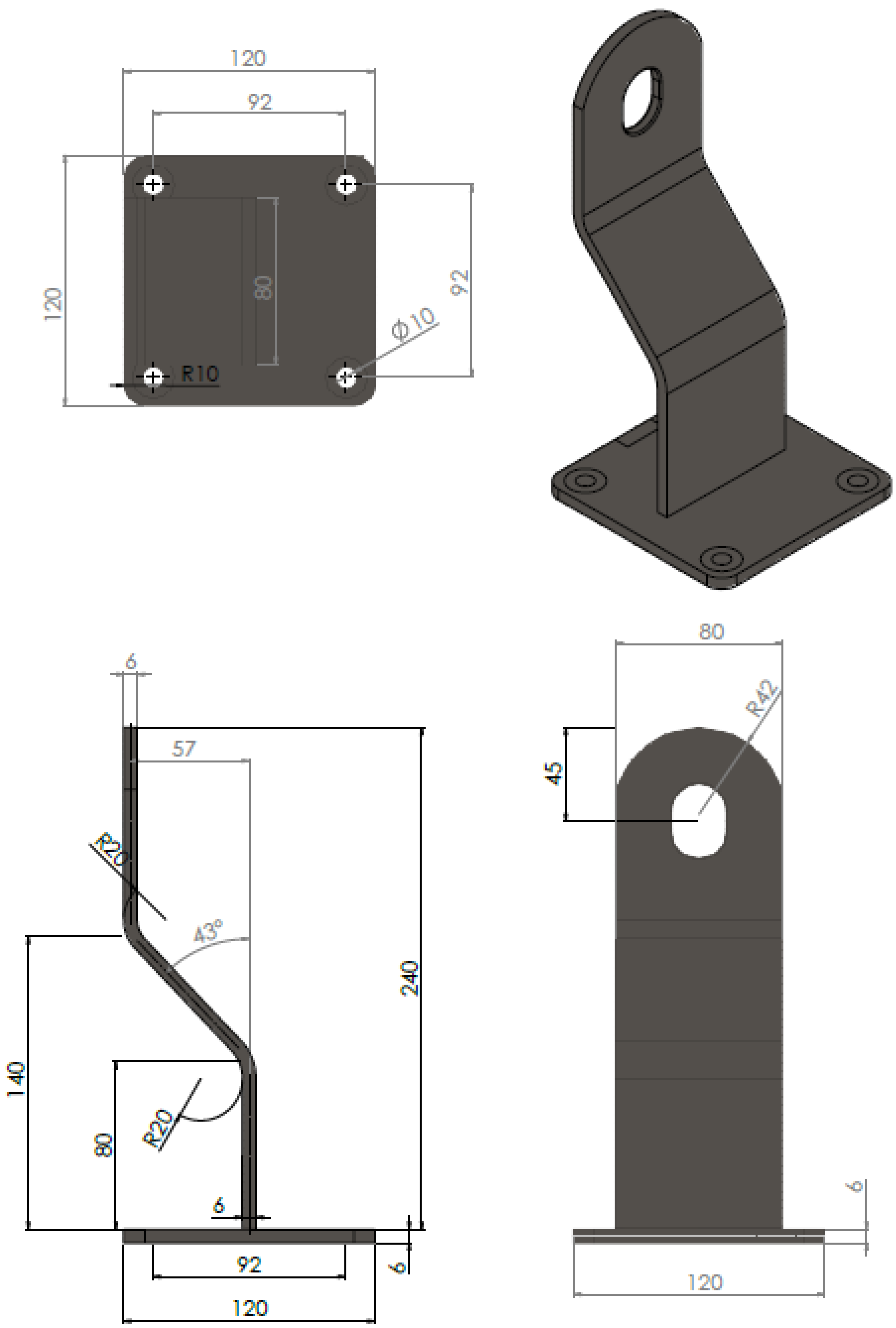
5.0 Resources Required

To implement this best practice, the following resources are required:

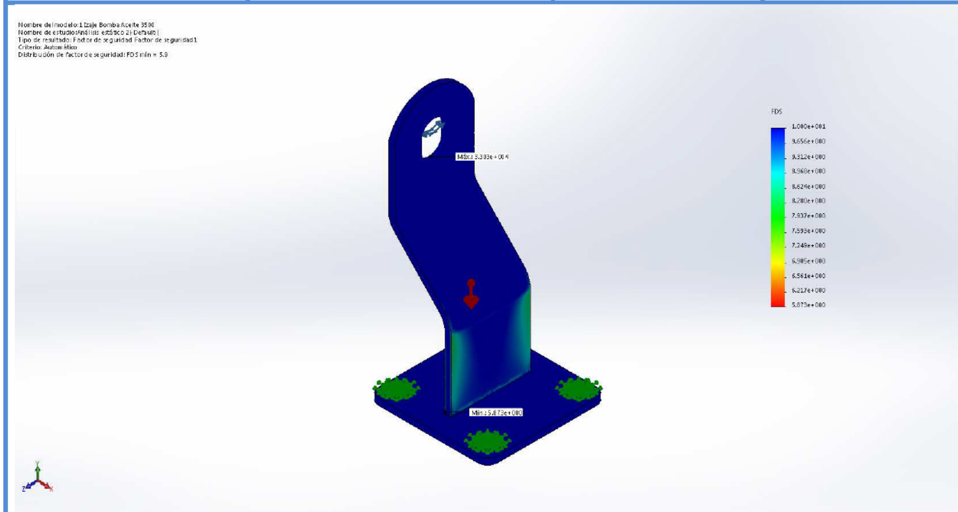
- Materials as outlined in the *Best Practice Description* section
- One welder technician

6.0 Supporting Attachments / References

Below are the required manufacturing drawings and the stress analysis conducted on the lifting tool.



Nombre	Tipo	Mín.	Máx.
Factor de seguridad1	Automático	5.87304 Nodo: 96013	33031.1 Nodo: 98350



1 Izaje Bomba Aceite 3500-Análisis estático 2-Factor de seguridad-Factor de seguridad1

FEA analysis, FS = 5.9

7.0 Related Best Practices

No specific lifting tool is provided on SISWeb. Refer to SENR1126 for information on the Oil Pump installation.

8.0 Acknowledgements

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